European Commission

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Notional Nozzle Models for Hydrogen Releases from High Pressure Systems

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Joint Research Centre

European Commission

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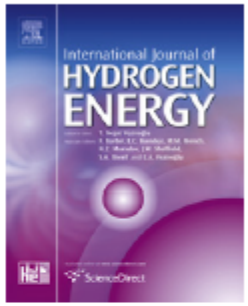

Evaluation of notional nozzle approaches for CFD simulations of free-shear under-expanded hydrogen jets

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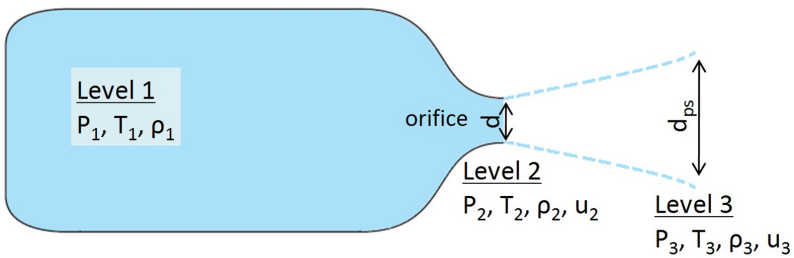
Notional Nozzle Concept

Need for reasonable computer run-times, for engineering applications

Replacement of the actual release nozzle by a notional one



Under-expanded jet release

Level 1: "reservoir" conditions

Level 2: real orifice

Level 3: notional nozzle location

The notional nozzle does not exist in real jets since it is based on strong simplifications/assumptions:

- uniform velocity profile
- uniform concentration profile
- flow properties from conservation equations applied on the jet centreline

5 Notional Nozzle Models

- **Birch et al. (1984):** Conservation of mass, temperature at notional nozzle equal to atmospheric
- **Birch et al. (1987):** Conservation of mass, conservation of momentum, temperature at notional nozzle equal to atmospheric
- **Ewan and Moodie (1986):** Conservation of mass, temperature at notional nozzle equal to the one at the actual release nozzle
- **Schefer et al. (2007):** Conservation of mass, conservation of momentum, real gas (Abel-Noble equation of state), temperature at notional nozzle equal to atmospheric
- **Harstad and Bellan (2006):** Conservation of mass, conservation of momentum, conservation of energy, location just after the Mach disk (Low flow speed).

3 Turbulence Models

- Standard k-ε
- Shear Stress Transport (SST)
- Baseline (BSL) k-ω model

RNG model was abandoned because grid independence could not be reached

	Coarse	Fine	**Finer**	Finest
Total number of nodes	47.500	68.800	179.300	533.300
Number of nodes in the conical geometry	5.500	19.300	72.100	344.600
Minimum (10^{-4} m) and maximum (10^{-1} m) grid size in the conical geometry	4.8 1.56	4.8 1.1	4.8 0.69	4.8 0.3

FZK/KIT Experiments

Simulation Pressure (bar)
53.27
113.3 102.2
162.0

Test No.	Nozzle Diameter (mm)	Pressure (bar)	Flow rate (10^{-3} kg/s)	Distance from nozzle (m)
HD35-37	1	54.6	2.46	0.75
		52.1	2.35	1.5
		53.1	2.39	2.25
HD00-02	0.75	113.3	2.87	0.75
		104.8	2.66	1.5
		99.6	2.53	2.25
HD22-24	0.25	162.8	0.46	0.75
		160.4	0.45	1.5
		162.1	0.46	2.25

Experimental conditions

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Conditions at the Notional Nozzle

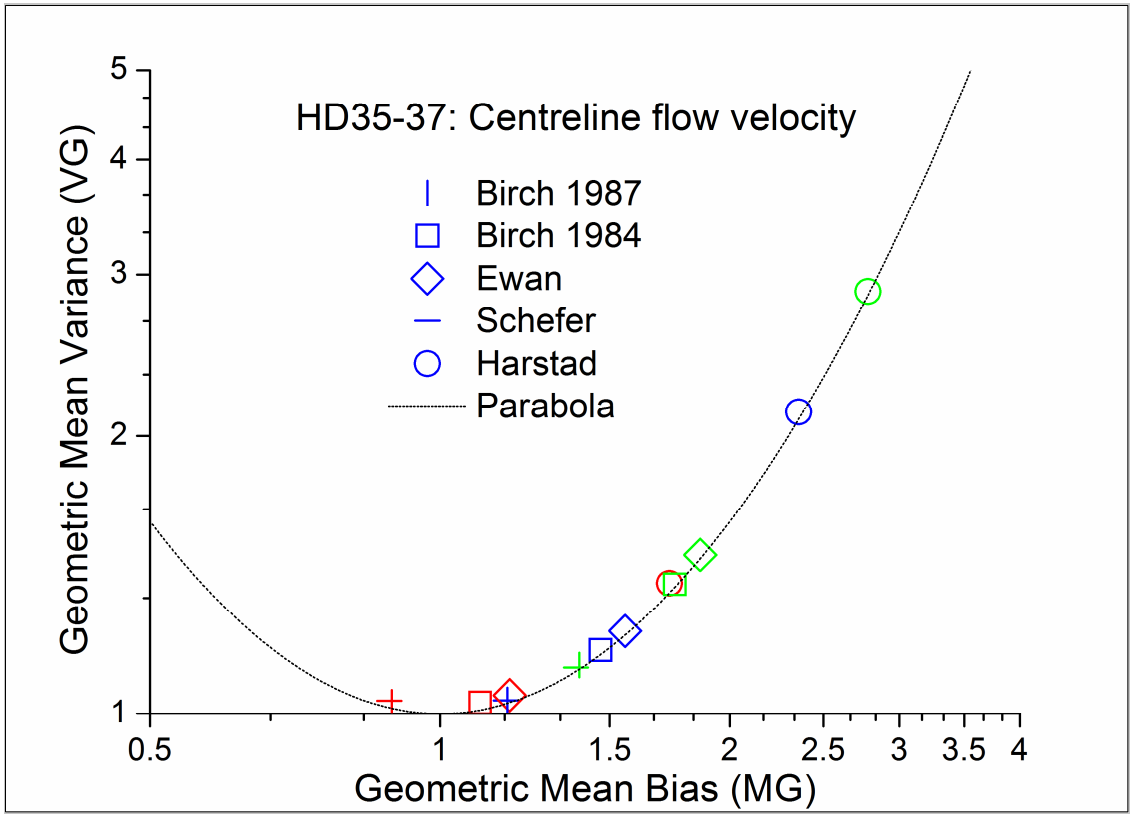
Approach	Temperature (K)	Density (kg/m^3)	Velocity (m/s)	Diameter (10^{-3} m)	Mass flow rate (10^{-3} kg/s)
	HD35-37 (1 mm nozzle diameter, 53.27 bar)				
Birch1984	287.6	0.0854	1292	5.262	2.4
Birch1987	287.6	0.0854	1984	4.247	2.4
Ewan	238.8	0.1029	1178	5.024	2.4
Schefer	287.6	0.0854	2001	4.228	2.4
Harstad	278.4	0.0882	510	8.241	2.4
Exp. data	237	2.74	1200	1.0	2.4
	HD22-24 (0.25 mm nozzle diameter, 162 bar)				
Birch1984	287.6	0.0854	1292	2.303	0.46
Birch1987	287.6	0.0854	2004	1.849	0.46
Ewan	238.8	0.1028	1178	2.199	0.46
Schefer	287.6	0.0854	2055	1.827	0.46
Harstad	278.7	0.0881	500	3.645	0.46
Exp. data	234	7.76	1239	0.25	0.46

Experiment/Simulation comparison with statistical analysis

$$MG = \exp\left(\overline{\ln V_o} - \overline{\ln V_p}\right) \qquad VG = \exp\left[\overline{\left(\ln V_o - \ln V_p\right)^2}\right]$$

- Geometric Mean Bias (MG): it measures relative mean bias and indicates only systematic errors based on a logarithmic scale.
- Geometric Mean Variance (VG): it measures relative scatter.
- Vo is the observed (experimental) value and Vp is the predicted (numerical) value

HD35-37: 1 mm nozzle; 53 bar

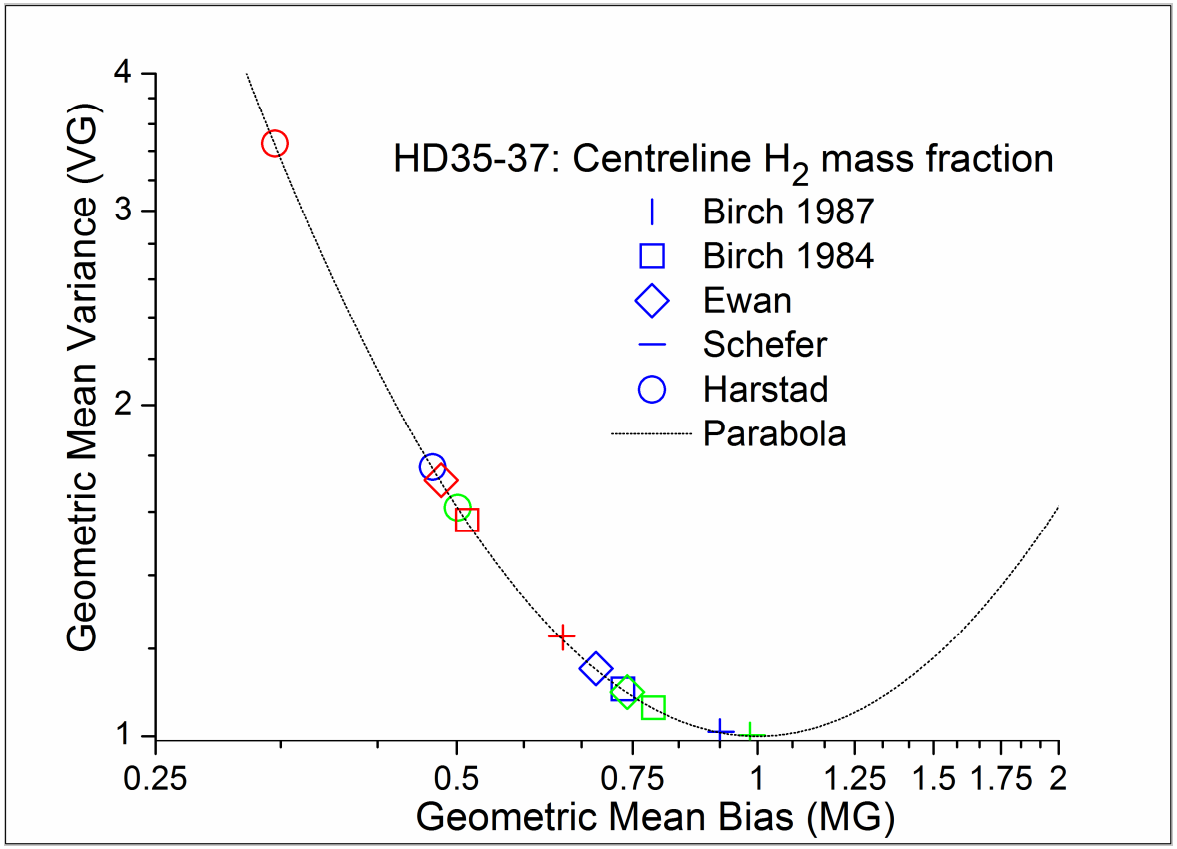


Green →SST
Blue→BSL
Red→k-ε

MG<1 over-prediction

MG>1 **under**-prediction

HD35-37: 1 mm nozzle; 53 bar



Green →SST
Blue→BSL
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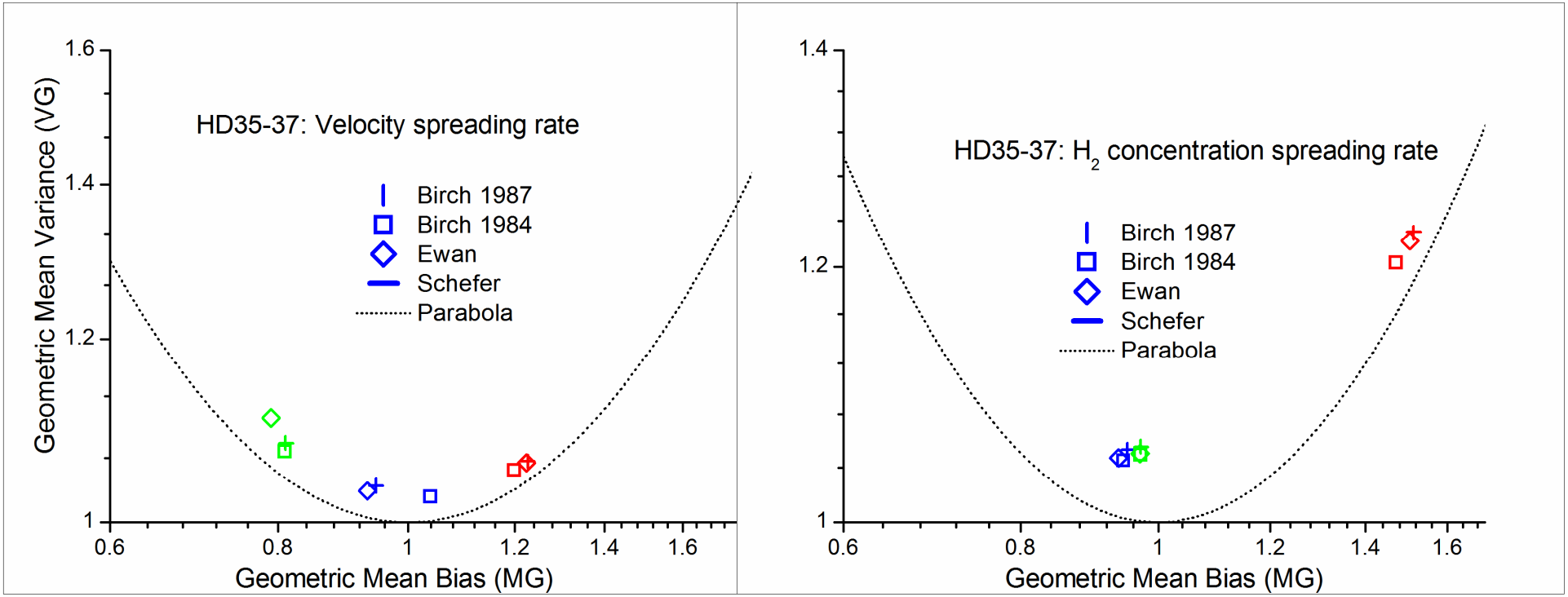
MG<1 **over**-prediction

MG>1 under-prediction

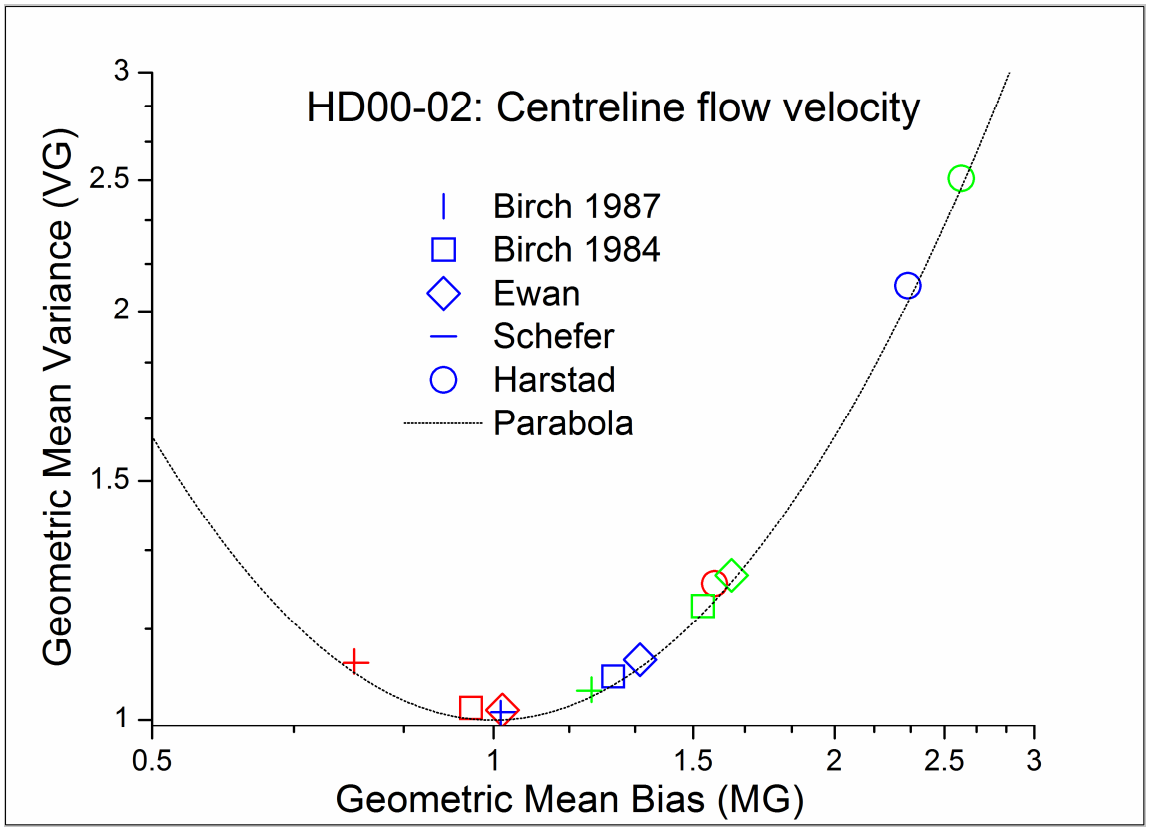
HD35-37: 1 mm nozzle; 53 bar

Spreading rates: the value of the radial distance from the centreline where H2 mass fraction and velocity are half their centreline value.

(Green →SST Blue→BSL Red→k-ε)



HD00-02: 0.75 mm nozzle, ~100-112 bar



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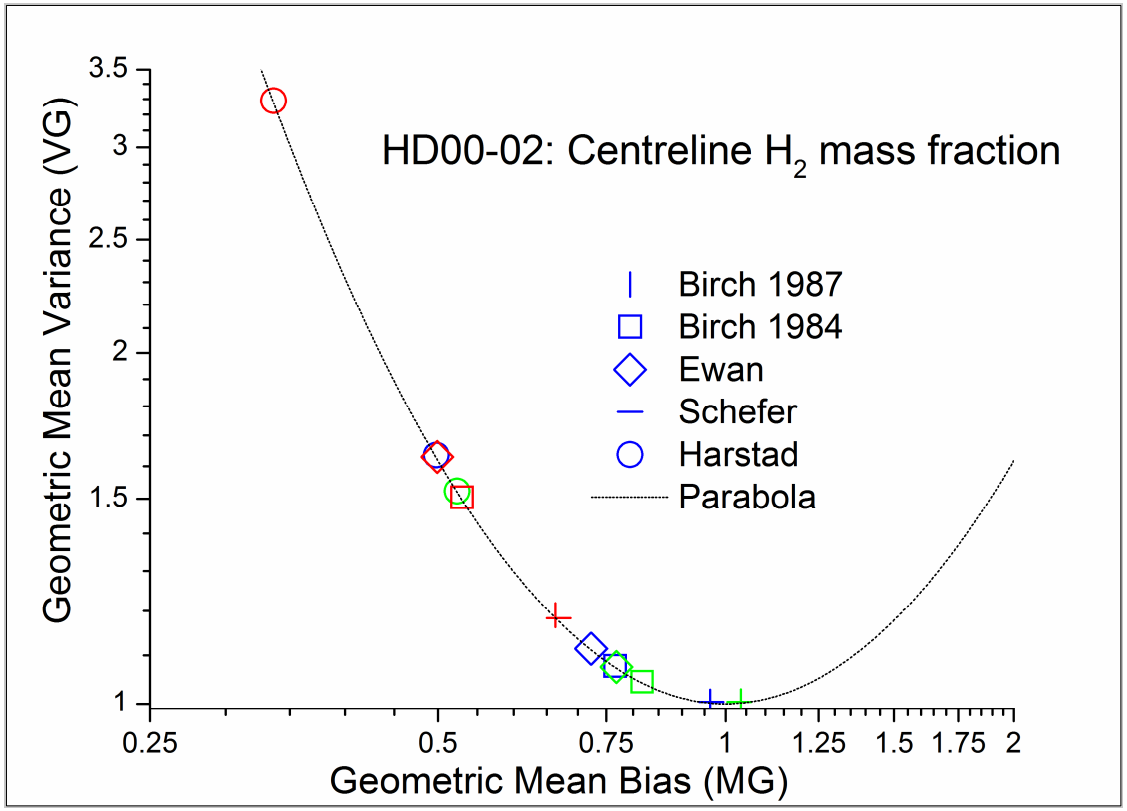

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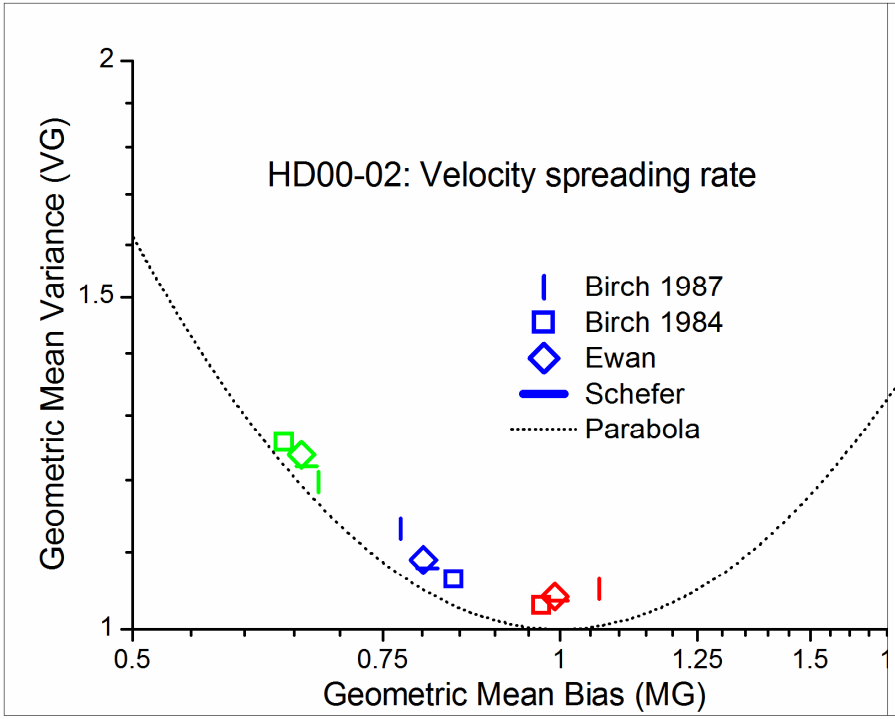
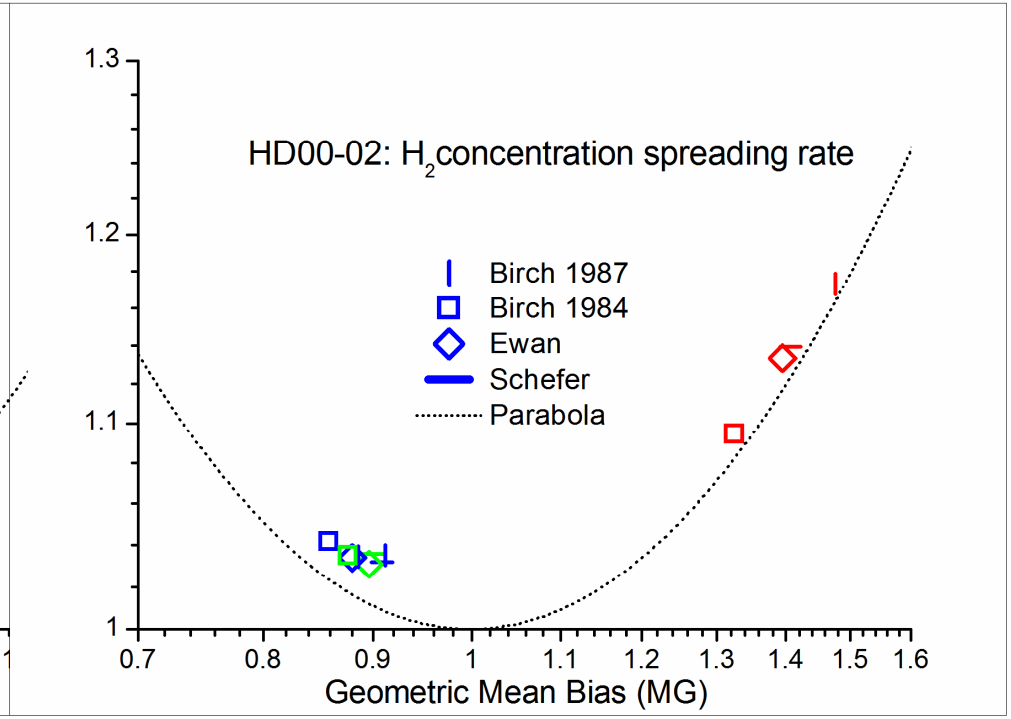
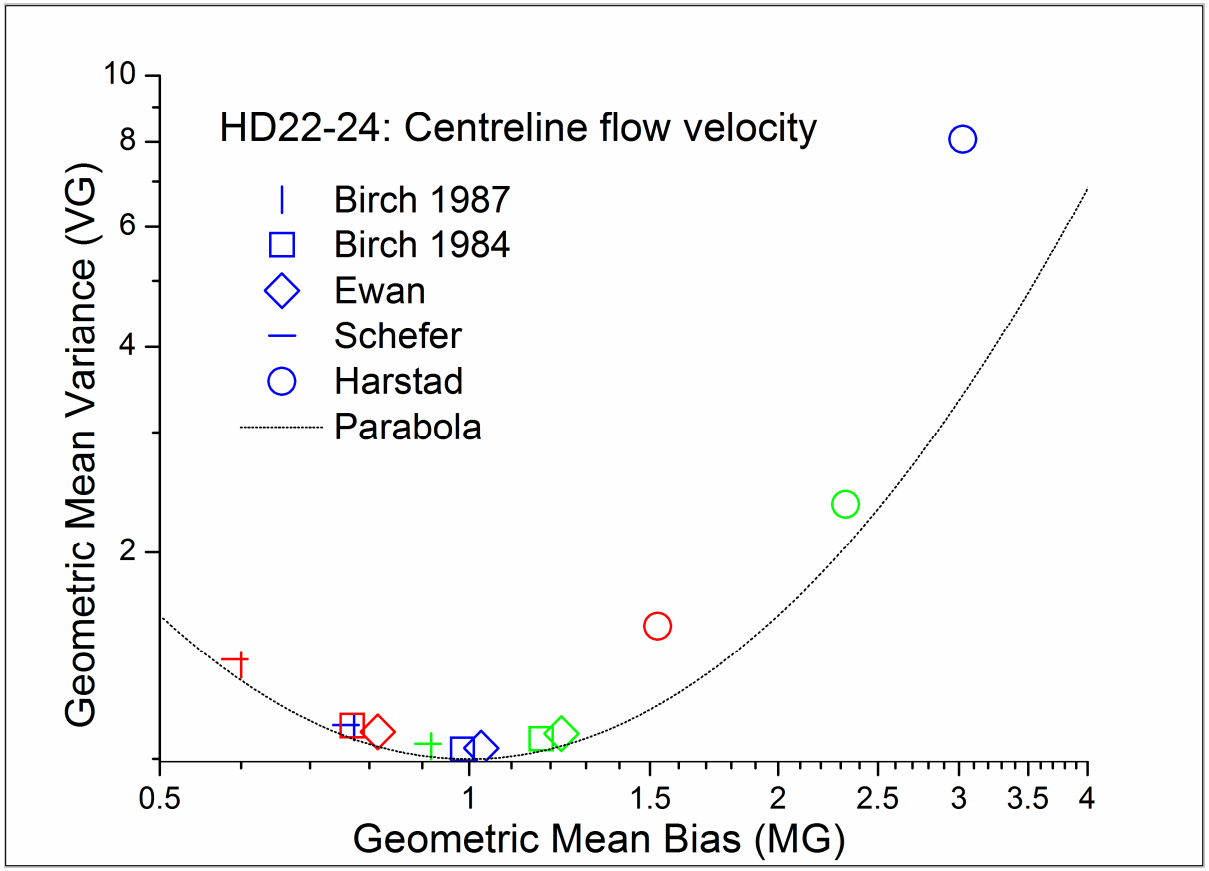




HD22-24: 0.25 mm nozzle, ~161 bar



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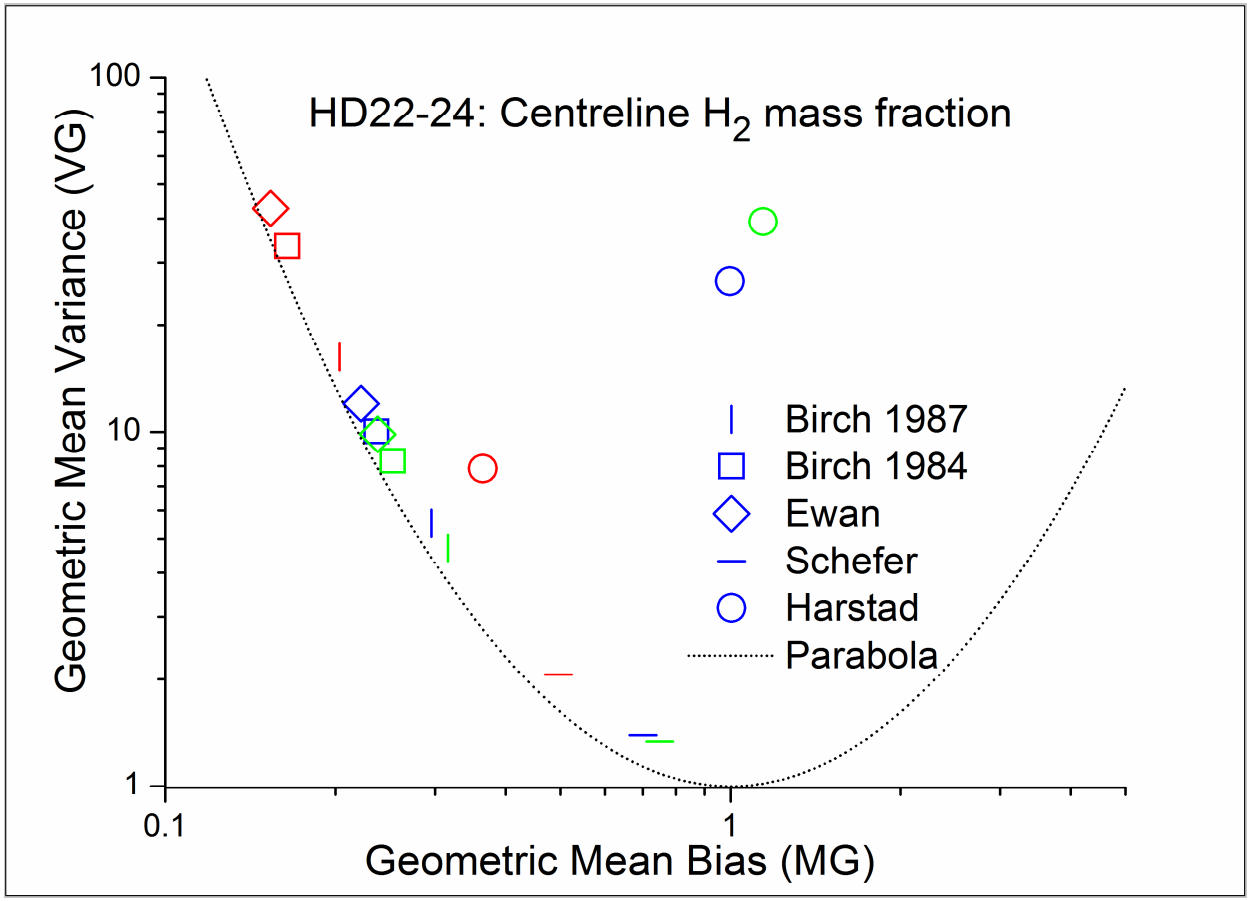

Green →SST
Blue→BSL
Red→k-ε

In general simulation results are less accurate than at lower pressure and with larger diameter.

By comparison of Schefer and Birch87: need for real gas equation at high pressure

Identified trends

For higher pressure and smaller diameter, the accuracy decreases.

Accuracy of the notional nozzle models in the following order, given the same turbulence model: Birch87/Schefer, Birch84/Ewan, Harstad.

Accuracy of turbulence models. For the concentration, SST and BSL perform in general better than k-ε. For the flow velocity, the results give less uniform indications. In some cases, k-εis the most accurate model.

Those conclusions are valid for the 3 cases that have been considered in the investigation and for the selected approach (e.g. unstructured mesh). Other cases should be considered to give more general validity to the conclusions e.g. very high pressure (500-700 bar), different diameter size, jet impingement with obstacles/walls.

GAPS

Validation of more recent notional nozzle models e.g. Molkov's

Validation with comparison for other extremely relevant parameter for the combustion e.g. not only concentration and flow speed but also turbulence.

Validation with broader range of conditions: very high pressure (500-700 bar), different diameter size, jet impingement with obstacles/walls.

Definition of criteria to help to identify the best modelling strategy for each application: which notional nozzle? notional nozzle or real nozzle?